

Week 12 Notes

1. Plan:

1) **Functional-Substitutional Methodology:**
Frege, Bolzano, Ampliativity, and the Role of Logic

Fregano again:

Ampliativity. Inferential and Constructional (*salva consequentia* and *salva veritate*).

What Frege did that Bolzano did not, and *vice versa*.

Kant learned two lessons from his study of Euclid:

1. People thought that **the lesson of Euclid** for philosophers **lay in the *proofs***—that is, in the *coercive* nature of *reasons* manifested in proofs.

But Kant had the idea that it lay in the fact that you could infer (prove) *new* things.

Euclidean ampliative inferences vs. syllogistic single premise ones.

Frege in effect saw (Macbeth taught us) that the secret to *that* lay in the way figures could play different roles in other figures, which determined what inferences and constructions could be made.

The lesson is that the Euclidean case works because one figure, e.g. a pair of points can serve multiple roles in other figures: as determining a line, as radius of a circle, as base of a triangle. The Macbeth point is that this is strictly analogous to Fregean expressions, which can serve as arguments to many functions, including being part of many function specifications. This is the idea behind the computational model of senses, as constellations of modes of presentation of *Bedeutungen*, which are the values of functions applied to arguments. The sense of an expression consists in all the ways it can be analyzed functionally-substitutionally into functions-applied-to-arguments.

These are *essentially multipremise* inferences.

Bolzano shows how to apply Frege's functional-substitutional methodology to make multipremise inferences intelligible.

Compare:

- a) Move to *polyadic* predicates requires Dummettian complex predicates, for Frege derived by functional-substitutional analysis.
- b) Those polyadic predicates are involved in *essentially multipremise* implications and incompatibilities.

- c) We saw that something new happens when *alternating quantifiers* are applied to *relational* (polyadic) complex predicates. This is the Hintikka-Friedman understanding of the roots of Kant's concept of synthesis in his analysis of Euclid.

This is *constructive, logical ampliativity*.

2. A kind of ***pragmatism***:

- a) **Euclid** starts with two kinds of **primitive abilities**: given any two points, to draw a line, and given any two points, to draw a circle with one as center and the other marking the radius. He then shows how to ***elaborate*** those *primitive practical abilities* so as to build the capacities for instance to draw *triangles*, pick out *midpoints* of line segments, distinguish and compare *angles*, generate *similar* triangles, and much more.
- b) For many of these, the elaboration is done by **straight-schedule algorithms**: mere lists of steps. Later on, he appeals to **conditional branched-schedule algorithms**, where the steps depend on the results of previous steps (do the lines intersect? Does the line intersect the circle at two points, one point, or not at all?...).
- c) Computation should be thought of as part of *pragmatics*: an account of what one must (be able to) *do* in order to be thereby manipulating tokens with a certain kind of content.
- d) A ***computational pragmatics*** would start with primitive abilities, and exhibit the use of expressions as the exercise of abilities algorithmically elaboratable from those primitive abilities.
- e) Kant is a pragmatist—though not *only* that.

For he looks at what subjects must *do* in order thereby to be manipulating concepts and judgeables: *synthesize* a a transcendental unity of apperception. He learned this from Euclid.

Put otherwise: **pragmatism is just the American version of neoKantianism.**

It is different from **German neoKantianism**, which focused on the **normativity** of discursiveness, in that it focuses on the practical activity of *using* concepts (to make judgments, so, on *judging*) to understand the *contents* of those judgments.

Pragmatist neoKantianism focuses on what one must *do* in order to count thereby as judging, thinking, or representing anythings.

3. **Frege is Kantian** in looking to features of *representings* to infer features of *representeds*. So, in the *GL* he understands the question “Are numbers objects?” as meaning “Are numerals proper singular terms?”.

The essence of the conceptual phenomenalism that is an important aspect of transcendental idealism is looking to what we have to *do* in order to judge (namely *combine, synthesize* different representations) to understand what we are making judgments *about*.

This Kantianism comes out in his **taking the realm of *Bedeutungen* to coincide with *reality***: subdividing it into his three realms.

4. Frege has more to say about *pragmatics*—the study of the *use* of linguistic expressions—than is generally recognized.

- a) His account of *judging* (and its linguistic expression: *asserting*) is that it is “the recognition of an expression as (a name of the) True.”

All judging is accordingly assimilated to identity claims that have “the True” flanking the identity sign.

This tells us about their *inferential role* via the account of identity claims as intersubstitution licenses.

The picture is that sometimes one looks at an expression and recognizes it as a name of one’s old friend (implicitly familiar to everyone who can judge), *the True*—as another *guise* that that familiar object shows up as or in.

- b) Further, he implicitly has (gives us all the conceptual raw materials needed to supply) **an account of what *grasping a sense* (paradigmatically, a *thought*) consists in.** It is grasping its *inferential role* (*begrifflich Inhalt*).

And he has an account of what one must do (in a perspicuous notation) in order to do that. It is being able to break it down in a function-argument way (so, substitutionally) so that one knows at least how the *Bedeutung* of the expression can be computed.

- c) The target ability, grasp of a thought, is practical mastery of its inferential role, primarily, its role as a *premise* in *multipremise* implications (and incompatibilities), as defined in §3 of *BgS*, in the definition of *begrifflich Inhalt*.

- d) **That practical ability to distinguish good from bad implications in which the sentence figures is in turn broken down into the capacity to discern the f-a-t-a s contained in the thought, and so practically *know how* (= be able) to compute the *Bedeutungen* those f-a-t-a s are modes of presentation of.** That lets one determine which substitutions preserve *truth*, which determines a class of *implications* (reason relations).

- e) **In general, what one needs to be able to *do* in order grasp a sense, to begin with, a sentential sense or thought, is to determine its inferential role *by* computing its *Bedeutung*.**

- f) By *GG*, all inferences are licensed by identity claims, construed as intersubstitution licenses.

4. Frege and Bolzano:

- a) There is the Bolzano method for understanding essentially multipremise reason relations substitutionally, so analytically, quite apart from any logical vocabulary.

For Bolzano, an implication is good just in case *there is a partition* of the whole vocabulary into fixed/variable, such that *every variation* of the variable vocabulary (substituting some *other* variable vocabulary for it) that makes the *premises* true also makes the *conclusion* true.

b) For Bolzano, the only thing special about *logical* vocabulary (as the fixed vocabulary) is that it is easier to figure out which implications are truth-preserving under substitution. He is thinking exclusively analytically, so top-down. And he does include sentences with iterated quantifiers as among those whose implications/incompatibilities are to be determined by his definitions. What he does *not* do is formulate *rules* letting us compute what implications are good. He does not offer, as Frege does, a *proof theory* for logical vocabulary. In effect, he only gives us a *semantics*. He defines, by substitutional equivalences, which implications/incompatibilities *are* good, without telling us how to find the good ones.

b) Bolzano treats all vocabulary, all partitions into fixed/variable, as on a par *semantically*, finding differences among them only *epistemologically* and *heuristically*. By contrast, Frege *only* addresses *logical* vocabulary.

Along this dimension, Bolzano is *much* more general in his semantic concerns than Frege is.

c) But Frege has an important idea that Bolzano does not: **logical expressivism**. Frege has the idea that what is special about logical vocabulary is that it can *make explicit* the implications/incompatibilities that govern *nonlogical* vocabularies: the culinary, nautical, theological, geological, and so on partitions into fixed/variable that are Bolzano's topic. Considered from the point of view of its *expressive power*, *logical* vocabulary is *primus inter pares*. This logical expressivism is Frege's rationale for privileging logical vocabulary in a way that Bolzano does not.

Frege does determine which essentially multipremise implications/incompatibilities hold in virtue of logic alone. This is the application of Bolzano's method of determining consequences (and incompatibilities) applied to *one* way of partitioning vocabulary into fixed/variable, namely fixing the *logical* vocabulary.

But logical vocabulary *also* can *express explicitly* the implications/incompatibilities of any of the *other* Bolzanian senses of 'consequence'.

This will take the form of *theories*, consisting of logically complex claims, involving conditionals and negations (and the aggregative connectives).

d) So I am seeing Frege as going far beyond Bolzano along two dimensions:

- i. **Logical expressivism**, privileging logical vocabulary by its expressive power to make explicit the reason relations of implication and incompatibility that articulate the *conceptual contents* (*begriffliche Inhalten*) of *all* vocabularies, and
- ii. A proof theory that lets one *compute* the implications and incompatibilities that hold in virtue of logic alone.

We can think of these moves as layered on, deepening, and developing Bolzano's broad vision of multipremise implications..

e) And there is also a variety of purely *logical* techniques for producing ampliative multipremise reason relations. (The kite.)

The kite is of specifically *logical* novelty. So Bolzano has no place in it. But is it term definition or multipremise inference? Complex predicates requires polyadicity, but essentially involves multipremise inference (does it?). Alternating quantifiers is about both. Euclid-Skolem function story is about constructing or computing objects, in order to determine inferences. So what is made explicit by these alternating quantifiers is a crucial point of connection or interaction **between the term-definitional and the multipremise consequential senses of novelty or ampliativity**.

- f) Theme of ampliativity is about construction in the sense of defining new things (and then, secondarily, their inferential relations) from old ones.

It is not *proving* things, but *defining* things that matters most. (Cf. Kant on Euclid, above.)

In this sense, *construction* first, then *inference*.

As per GL, new sortal and singular terms from old.

A deep issue is how such constructions are related to the fact that multipremise inferences can yield new conclusions. That last is the inferential version of the constructional-definitional.

The connection between them is substitutional, since identities are intersubstitution licences (a pragmatic fact), and intersubstitutions license or preserve inferences.

Should use this **constructional-definitional vs. consequential-inferential division**, with twin issues of ampliativity (defining new—in various senses—terms, sortal and singular, paradigmatically for Frege by abstraction, from old, and *multipremise* sets containing collectively conclusions their premises individually do not, which can be understood functionally, or, in a better further use of Bolzano's application of substitution to consequences rather than just sentences) to structure my recollective reconstruction.

Bolzano takes us from objects (in the Fregean sense, truth values, which is not B's sense of object) **to concepts as inferential roles**. He forges a strong relation between the two orders: truth/consequence. And he does that using the very same substitutional methodology Frege uses.

This is a way to integrate **the semantic novelty-ampliativity theme**.

Now, **divided and related as terms-definitions-constructions vs. sentences-consequences, including abstraction** and logical kite related to multipremise reason relations.

A big theme should be the combination of function/argument substitutional analysis (including the huge generalization of the notion of function, to arbitrary arguments and values), and using intersubstitutability *salva* something as the equivalence relation that is an abstractor for the introduction of new terms: both sortal kind terms and singular terms falling under them.

Substitution (variation) is a species of abstraction, not the genus. This theme includes Frege's extension of the concept function to include all substitutionally distinguishable expressions (and then, their *Bedeutungen*).

Set theory elaborates the member-of relation,

category theory elaborates the composition of mappings relation, and
homotopy type theory elaborates generalized notions of identity.

g) Going back and forth between concepts and objects (sentences and terms).

Constructing objects from concepts, by abstraction, and objects to concepts by forming functions by substituting for objects (terms).

Concepts extensionally, as mappings from terms to sentences (on truth = Bedeutung side) and intensionally, as mappings from terms to sentences (on consequence = sense side).

Concepts as sense functions (intensionally): functions from term senses (computationally relate to Bedeutungen via modes of presentation) to **thoughts** = sentence senses = inferential roles (individuated by intersubstitutability *salva consequentia*).

Thoughts : truth values :: *salva consequentia* : *salva veritate*.

These are the two, paired, semantic substitutional hierarchies.

2) Abstraction:

For singular term use, *GL* teaches

- i) The use of singular terms is governed by identity claims.
- ii) Identity claims are *recognition judgments* (judgeables), which “express the recognition of an object as the same again.”
- iii) Identity claims are intersubstitution licenses (*salva veritate*, in some cases, *salva consequentia*).
- iv) There must be a governing sortal kind term for each singular term.
It supplies the criteria of identity and individuation, which determine the truth of identity claims.
- v) If other singular terms governed by a sortal are already in use, the identity/non-identity of any new singular terms governed by that sortal must be settled.
The identities with singular terms falling under *other* sortals will already have been settled, and so can be inherited by the new ones.
- vi) Recall argument that *strongly cross-sortal* identity claims are almost always false, because of the modal involvement of terms: the fact that they have subjunctively robust properties, and substitution licenses apply to them, too.

Overall theme is *abstraction*.

- a) Rehearse the *GL* definition and analysis.
An equivalence relation (reflexive, symmetric, and transitive) in the *old* vocabulary is turned into or treated as an *identity* statement in a *new* vocabulary.
This introduces new *sortal kind terms* and *singular terms* falling under them (e.g. directions of lines).
- b) Issue: the Julius Caesar problem. Cannot assume that because a term is *introduced* by abstraction that what is introduced is a *kind of thing*.
This is a **semantic version of the Platonic Principle, construing epistemological differences as ontological differences**.
- c) Frege is absolutely committed to not all terms introduced by abstraction being of an ontological kind abstract objects. For *Bedeutungen* can all be introduced by abstraction, but are coreferential with (identical to the objects of) ordinary empirical descriptive (OED) terms:
***Bedeutung*(‘Fido’) = Fido.**
- d) For numbers, the **third *GL* definition invokes extensions**, which later, in “Function and Concept”, are *courses of values*, carried on in *GG* by Axiom V.
- e) Problem there: too much expressive power in forming *objects* from *concepts*. Result is inconsistency. Move from logical concepts to logical objects raises issues, as in (22) above.

- f) All logical objects in *GG* are courses of values.
But not all terms introduced by abstraction are courses of values. Though they can be horsed around into that form if we want, we need not do so. For instance: directions of lines.
- g) And the fact that there are *abstracta* that are not courses of values matters for understanding Frege. For *all* his basic semantic-logical concepts are introduced by abstraction: *begriffliche Inhalten, Bedeutungen, Sinne, Wertverläufe*.... The Julius Caesar problem is particularly acute for *Bedeutungen*, via the principle that the *Bedeutungen* of ordinary singular terms are the objects named, referred to, or picked out by them, the principle that **Bedeutung($\ulcorner t \urcorner$)=t**.
(Name for this principle? The name-bearer model.)

Problems with abstraction:

- a) **Objection, from results in *GG*: This definition is too expressively powerful. It leads to inconsistency by allowing definition of inconsistent logical objects.**
(Inconsistent logical *concepts*, or inconsistent *senses*, is OK, but not singular term *Bedeutungen* whose existence one can ‘prove’.)
- b) Reiterate that one can understand the troubles of *GG* at a very abstract level like this:
Inconsistent concepts are not a problem. It is true that (necessarily) nothing falls under them, they are true-of no objects. But that is true of lots of concepts, and argument *ad absurdum* depends on the legitimacy of such concepts.
But inconsistent *objects*, in the sense of objects that have inconsistent *properties*, so that incompatible concepts are true of them, are another matter. So **the move in *GG* to introduce courses of values, as logical objects, from the various logical concepts defined, is a decisive, potentially disastrous, step.** Note that in doing this, Frege is not only fixing or cleaning up the notorious footnote about extensions at the end of *GL*, (and using the apparatus of terms, identity, substitution there), but also showing how to think about “Concept and Object”. There *are* objects we can call “the concept horse”, but they are its course of values. (It remains a problem that Frege himself needs to talk about concepts, and to distinguish the concept horse from its course of values.). These are all threads from earlier works that come together in this last major work, and which I should rehearse in my Conclusion.
- c) Should use the discussion of **dangers of move from logical concepts to logical objects** to introduce second point: that another problem—and potential direction for a solution—is not to require settling the truth of *all* identities, so long as it is settled *which* identities, and so implications, *are* settled. This is the “Significance of Complex Numbers...” point.

This can all be used to move on to the other material on abstraction.

- d) Is the problem instead perhaps (or in addition?) requiring settling truth of *all* identity claims involving the new vocabulary? Can it be enough to know that some identities are not settled either way, rather than insisting on fixing all the heterogeneous identities as *false*?

- e) Relaxing Frege's requirements:

Frege requires that *all* the possible identities have truth-values associated with them (and then, exclusively and exhaustively).

This gets him into trouble with courses of values.

But his inferentialism does not require it.

Having a definite inferential role is compatible with some of the truth-values of identities being indeterminate—as long as we can specify which those are.

More generally, can settle some implications as good and others not, even if not all the identities are settled.

- f) Frege believes that in order to settle *definite* or *determinate* inferential roles, one must settle the truth of *all* the identities in which each singular term can occur. That is the origin of the “sharpness of boundaries” of concepts requirement—because it is intersubstitution as arguments of functions (f-a-t-a s as components of senses) that are licensed by identity claims.
- g) It can be doubted whether the connection Frege insists on is the form of only connection possible between *determinate inferential role* (in what sense of ‘determinate’?), which is an independently motivatable requirement, on the one hand, and the *truth of intersubstitution-licensing identity claims*, on the other. The latter depends on
 - i) a range of functional contexts, which could be wider or narrower, while still settling *determinate* inferential roles, and
 - ii) a range of possible substituends, which can flank the class of identity claims whose truth-value must be settled to settle which intersubstitutions are licensed (relative to the functions in (i)). Again, so long as which class of identity claims are licensed—doubly indexed by class of functions and class of substituends—is determinate, so will the inferential significance be, since that is just what intersubstitutions are licensed.

It seems that what matters is just that it be settled which intersubstitution licenses are being endorsed, for that settles what implications/incompatibilities hold inferentially. This is all “The **Significance of Complex Numbers** for Frege's Philosophy of Mathematics” stuff.

At least the last part should be (must be?) saved for next time (Week 12, the Conclusion).

Q: Can we do with less than fixing the sense of *all* the identities?

The argument about complex numbers and ‘schmingular terms’ suggests there are other stopping-places.

But giving up Frege’s “sharp boundaries” claim is giving up a *lot*.

We seemed forced to do so even for the mathematical case.

- 3) **Frege's 3-world ontology:** subjective/objective, with objective subdivided into *actual* (*wirklich*) and *non-actual*. Q: Where are (what is the status of) things introduced by abstraction?

Frege's Three Worlds:

- a) Frege invites us to think of the world as divided, exclusively, and perhaps exhaustively, into three ontological realms:
- i. Subjective events and episodes, which are not shared or shareable between subjects. They are reproducible *within* subjects (I can entertain the same idea-type twice, but you and I cannot share idea tokens. Can you and I have the same idea type? Perhaps yes, but it is not clear that we could *know* that we did. Frege does not have developed views here.)
 - Objective items, which can be the same for all subjects. There are two kinds of these:
 - ii. *Wirklich*, causally efficacious/effectable items. These paradigmatically include the *Bedeutungen* of OED vocabulary.
 - iii. *Senses* of expressions, shareable and graspable. This category includes *thoughts*, the senses of declarative sentences. And it includes *facts*, which are just *true thoughts*: "thoughts that are true." Thoughts, and so facts, can be *about* (have as their *Bedeutungen*) subjective or objective-*wirklich* things, as well as other thoughts (thinkables, not thinkings).

Note that *all* of these can be the *Bedeutungen* of expressions, and in that sense things we can talk *about* or *represent*. Senses are the *Bedeutungen* of indirect speech: talk about what others are

- b) We can ask: Where are *concepts* in this botanization?
They are *not* senses, and they are *not* merely subjective.
Cause is a concept.

The *wirklich* world is articulated by concepts.

Properties and relations are concepts, and the *wirklich* world exhibits them.

Frege is a *conceptual realist* in this sense: he takes the *actual* (= *wirklich*) world to be *conceptually structured*. The contrast here, of course, is with *Kant*, who is the paradigmatic *conceptual phenomenalist*, as a transcendental idealist.

- c) Concepts are *functions*. So at least some functions are in the actual, *wirklich* world or realm.

Concepts are functions whose values are truth values.

So are *truth values* in the *wirklich* world?

If some *functions* (in the form of *concepts*) are in the realm of actuality = *Wirklichkeit*, then must the *arguments* and *values* of those functions also be in that realm?

Note that there are also *sense functions*, which take, e.g., the senses of singular terms and return *thoughts* as values. They, presumably, are in the realm of sense.

- d) So, generally, what is the status of *functions* whose arguments and values are in some one of the three realms? Are they also in that realm?

And can there be **trans-world functions**: functions whose *arguments* are in *one* realm and whose *values* are in another? Senses *determine* *Bedeutungen*. To say that is presumably to invoke a kind of *function* whose arguments are senses and whose values are *Bedeutungen*, for instance dogs.

The computational model has senses *consisting* of functions, whose values are the *Bedeutung* of the expression of the sense.

What is the ontological status of those functions? They might take actual or *wirklich* items into *wirklich* items.

By making senses into constellations of functions from actual arguments to actual values, we are defining at least those senses entirely in terms of actual items plus functions relating them. Is that to place at least these basic senses of OED vocabulary in the actual world?

- e) It seems, then that we must extend the three-realm picture to include *functions* from items of any realm to items of any realm. And these functions will not themselves be in any realm. And for functions that are *concepts*, we have to ask where the truth-values are. Truth values are not senses. The senses of sentences are *thoughts*, and truth-values are the *Bedeutungen* of thoughts: a distinctive kind of *object*.

But surely they are not actual = efficacious either.

- f) Conclusion:

So it seems we can deduce the existence of at least two other kinds of things, in addition to what is in the three realms: *functions* (which can be trans-categorical, in having arguments drawn from one realm and values drawn from another), and *truth values*, which are not functions, but another kind of argument.

- g) And what about objects introduced by *abstraction*?

All *Bedeutungen* can be so introduced—as corresponding to equivalence-classes of expressions intersubstitutable *salve consequentia*. And some of those are identical to actual, subjective, or *sinnlich* items, since *everything* (every *thing*, object, particular, non-function) can be the *Bedeutung* of *some* expression.

I would say that the direction of the Earth's axis is actual and efficacious.

It can figure in causal explanations—though perhaps its role there can always be paraphrased in terms of something that is *not* introduced by abstraction.

Certainly mass and force are—at least, masses and forces are.

Are the *concepts* of mass and force actual = efficacious?

The truth values are introduced by abstraction.

And so are *senses*.

It seems there can be *abstracta* corresponding to things of all three ontological domains.

Indeed, the *concepts* object and function are introduced by abstraction.

- h) Perhaps Frege's "third world" contains senses, but is not limited to them. Perhaps functions and (so) concepts, and *abstracta* that are *not* identical to actual or subjective things are *also* denizens of **the third world, which consists of *all* objective, nonactual things**. It is a motley. But so are the other realms.

One focus should be on the denizens of the third world: objective, rather than subjective, an unwirklich, rather than wirklich.

And one strand of that discussion is about the existence of *objects* (e.g. senses, particularly, thoughts).

Frege approaches that through *singular terms*.

The overall claim is that if a properly introduced (or used) singular term

- i) Occurs *essentially* in
- ii) A *true* claimable (thought),

Then

- iii) It refers to an *object* (particular) that *exists*.

A term occurs *essentially* in a thought in case

- a) it occurs in the sentence, in the sense that it can be substituted for, and
- b) some substituends for it make the claim *true* and other substituends for it make the claim *false*.

The paradigm for Frege is the *GL* argument that numerals are singular terms and they occur essentially in true statements of number ("The number of moons of Jupiter is 4,"), *so* natural numbers exist—there are natural numbers. They are *objective* but not *wirklich*.

This is a Kantian strategy. For Kant argues from features of the *representings* to features of the *representeds*. For instance, he argues from features of intuitions and concepts (e.g. their being products of receptivity or spontaneity) to the particularity or generality of what they represent, and from features of the activity of *judging* to features of the *judgeables*—here, from what one is *doing* in synthesizing a unity that has the structure distinctive of apperception to what kinds of *contents* judgeables must have: they must stand to one another in reason relations of implication and incompatibility.

- 4) **Sellars's 2-world ontology:** The world in the *narrow* sense is *discourse-independent*. The world in the *broad* sense includes discourse-dependent entities. Sellars's metalinguistic reading of *abstracta* leads him to exclude them from the world in the narrow sense. His inclusion of *propositions* (Fregean *thoughts*) as *abstracta* motivates his *nominalism*.

Against this rehearsal of the background, I want to add two new considerations:

Sellars (downstream from Frege's student Carnap—who wrote with Bruno Bauch because Frege was retiring) makes two moves:

- i. Distinguishing 'the world' in a *broad* sense and in a *narrow* sense.
What is "in the world" in the *narrow* sense is whatever could/would still be in the world in that sense if there were not, or never had been, any language users.
This is a counterfactual or subjunctive matter.
A paradigm is *prices*. Nothing would have a *price*, there would be no *prices*, if there were not *commodity exchange*. Absent such social practices or institutions, nothing can have the practical significance, or play the role of *money* or *prices*.
Whereas, there would still be rocks, trees, stars, photons.
- ii. Understanding *abstracta*, paradigmatically universals such as triangularity (note distinction between 'triangular' as a predicate and 'triangularity' as a property name, which gets smudged for colors—*can* use 'greenness', but usually use 'green' both for predicate and for property) as essentially *metalinguistic*.
Translation of "Triangularity is a property," as " 'Triangular' is a one-place predicate."

Sellars (addressing the issue of the ontological status of *abstracta*, for comparison with Frege):

Here I want to consider two distinctions, one Sellarsian, and the other my own Fregean distinction: the distinction between "the world in the *narrow* sense" and "the world in the *broad* sense," and the distinction between reference-dependence and sense-dependence.

- a) Where Frege has a *tripartite* or *triadic* ontological distinction of basic kinds of things (subjective, objective-actual, objective-nonactual), Sellars has a *dyadic* fundamental ontology: things in the world in the *narrow* sense and things in the world in the *broad* sense.

This distinction is *modal*, involving *subjunctive* (including *counterfactual*) properties. It turns on *language-dependence/independence*.

This is quite different from Frege's division.

- i. The world in the *narrow* sense is all the things that *could* still exist if there were not, never had been, and never would be language (and so, concept) users.

- ii. The world in the *broad* sense includes everything in the world in the narrow sense plus everything that is the product of discursive practices, including properties and relations that essentially involve linguistic expressions in use.
- b) Sellars takes *abstracta* to be specified in *essentially metalinguistic* vocabularies: vocabularies for talking about vocabularies.
 It follows, for him, that in worlds in which there are no vocabularies, nothing metalinguistic in this sense is true. There are not metalinguistic facts (= true claimables) in worlds that have no vocabularies being used, that have no discursive practices in them. So, abstracta do not exist in the world in the *narrow* sense: in the sense of **what would be here even if we weren't**.
- c) His point of departure is Carnap's analysis of the concept property.
 For Carnap, "Triangularity is a property," is a material-mode way of saying " '...is triangular' is a monadic predicate."
 Sellars saw that this won't do as stated, since "Triangularity is a property," says the same thing in English as "Dreieckigkeit ist ein Eigenschaft," says in German, and neither one is talking *about* terms in either language.
 Sellars can fix this up, with dot-quotes, while retaining the consequence that if there were no predicates at all, there would be no properties.
 For similar reasons, Sellars denies that propositions, and therefore *facts* (which, Frege-wise, he construes as true propositions) are "in the world in the *narrow* sense."
- d)
- e) t

5) Reference-dependence and sense-dependence of expressions.

Developing and applying a Fregean distinction:

Use Frege's key semantic concepts to distinguish *reference-dependence* from *sense-dependence* of expressions on expressions (so, an essentially metalinguistic distinction).

As part of this discussion, mention Kripke in *NN* as looking at dependence on *reference fixing* of expressions. This is what gives him *a priority* without *necessity*. It is a *kind* of sense-dependence: dependence on sense as what fixes reference, rather than sense as inferential role. It is sense as inferential role that, through its subjunctive robustnesses, incorporates *modal* information or commitments.

Then, I want to consider (a) in the light of (b). What is in the world in the narrow sense is what is *not reference-dependent* on what is specified in a *pragmatic metavocabulary*, which specifies discursive practices and the significances things can get by playing functional roles in such practices.

So we can take it that in general *Bedeutungen* of OED vocabulary are in the world in the *narrow* sense. The North Sea would be there even if we had no expressions to refer to it. But what about *senses*?

For Bolzano, thoughts or propositions are in the world in the narrow sense. They are in *that* sense 'objective', even though not *wirklich*.

For Sellars, propositions are metalinguistic, and for him it then follows that what they refer to depends for its existence on discursive practices. If there is no use of language, then no metalinguistic expressions have reference (*Bedeutung*). That is why he is an ontological *nominalist*: there are not facts in the world in the narrow sense, only things (objects, particulars—eventually, processes).

For Frege, facts are true thoughts.

So the question is whether *senses* (thoughts=propositions are senses of sentences) are objective in the sense of *not* being *reference-dependent* on discursive practices, say, not being reference dependent on the existence of *sentences* (that is, expressions *used as* sentences).

For Frege, we *grasp* senses by functionally decomposing *expressions*, in the end, *sentential* expressions (for without *sentences*, *truth-values* aren't in the picture, and without truth-values, *concepts* aren't in the picture). But the senses we grasp *are* supposed to be there, in the third world (objective, not *wirklich* = causally efficacious), independently of expressions for them. Thoughts are *sense-dependent* on discursive practices and expressions, but *not reference-dependent* on discursive practices.

2. I want to use a distinction that combines Sellars's *modal* distinction between language-dependent ("the world in the *broad* sense") and language-independent ("the world in the

narrow sense”) objects, properties, and relations, with the Fregean distinction between sense and reference. This is the distinction between *reference*-dependence and *sense*-dependence.

a) The distinction:

One vocabulary is *reference*-dependent on another if:

If none of the terms in the base vocabulary refer, then neither do the terms of the dependent vocabulary.

A world without siblings and parents is a world without aunts and uncles.

More controversially, a world without fluids is a world without pumps.

A world without masses is a world without forces. (Is this true?)

One vocabulary is *sense*-dependent on another if:

One can only *grasp the contents* expressed by one vocabulary if one *grasps the contents* expressed by another vocabulary.

b) There can be cases of sense-dependence without reference-dependence.

The example of response-dependent properties.

One form of *idealism* (“objective idealism”) and a corresponding form of *pragmatism*, starts off from the *sense*-dependence of all sorts of things on the concepts of a *pragmatic discursive metavocabulary*—the concepts that articulate discursive *doings* and *expressions* as playing roles in those discursive practices-and-abilities. A *bad* move from these considerations is to infer *reference*-dependence. Rorty’s (and Sellars’s) rejection of us-independent facts as “sentence-shaped bits of reality” is one manifestation of this tendency.

Abstracta are ‘metalinguistic’ in a sense that entails the *sense*-dependence of sortal and singular terms introduced by abstraction on the expressions for more concrete items appealed to. The analogy to *response-dependent* concepts is very close—and response-dependent concepts are my paradigm of sense-dependence without reference-dependence.

So I should lay out and then exploit this analogy:

Response-dependent concepts : concepts of responses

∴

Concepts introduced by abstraction : more concrete concepts appealed to in definition.

In both cases, the claim should be sense-dependence, not necessarily reference-dependence.

Sellars claims something stronger: abstracta are not in the world in the narrow sense *because* they are ‘metalinguistic’. This is the assertion of reference-dependence.

Q: As Sellars made a mistake here, due to not distinguishing sense-dependence from reference-dependence? Or does he have an additional argument? Or is he assuming further premises?

Sellars’s claim is that terms introduced by abstraction are *metalinguistic* in a sense that entails that their the *referents* cannot exist unless there are object-languages in use.

Sellars thinks that “grasp of a concept is always mastery of the use of a word,” and that *therefore* concepts are metalinguistic (the concept concept is a metalinguistic concept) in a sense that precludes them from being denizens of “the world in the *narrow* sense.” But concepts could be roles w/res to reason relations, and reason relations could have both objective and subjective (language-relative) forms.

Fregean concepts are not senses, but *Bedeutungen* of functions whose values are truth-values. Since they are not senses, they are not in the third world of senses. But does he think they are *wirklich*? For me, they can be. For copper is a concept, and being copper can *cause* the transmission of electricity.

Plan:

- 1) The Substitutional-functional methodology:
Frege, Bolzano, Kant and Euclid.
Ampliativity, and the Role of Logic
- 2) Abstraction. From concepts to objects.
- 3) Frege's 3-world ontology: subjective things, actual ('*wirklich*') objective things, and nonactual objective things.
- 4) Sellars's 2-world ontology: the world in the *narrow*, discourse-independent sense, and the world in the *broad*, discourse-dependent sense.
- 5) Distinguishing *reference-dependence* of one vocabulary on another from *sense-dependence* of one vocabulary on another.

Punchlines:

- 1) The **Substitutional-functional methodology**:
Frege, Bolzano, Kant and Euclid.
Ampliativity, and the Role of Logic
 - a) Bolzano's use of the functional-substitutional methodology is more general than Frege's along two dimensions:
 - i. Bolzano applies it to *define* the goodness of implications and incompatibilities generally (in terms of truth-values), *by* applying it to whole (pairs of) *sets* of sentences, not just to sentences and subsentential expressions.
 - ii. Bolzano applies it to *all* vocabularies, not just *logical* ones.But Frege pursues the project of *logical expressivism*: using logical vocabulary to *make explicit* the implications and incompatibilities
 - b) Frege follows Kant in aspiring to a *transcendental pragmatics* (Sellars's term), which aims at an account of necessary features of the *use* of *any* vocabulary.
Following Euclid, we can imagine this as a *computational pragmatics*, algorithmically elaborating a set of *basis abilities* into a set of *elaborated abilities*.
- 2) **Abstraction**:
Defining new *objects* from (Fregean) *concepts*.
All of Frege's technical concepts are introduced by abstraction: *begrifflich Inhalt*, *Sinn*, *Bedeutung*, *truth value*, *course of values*
Issue of cross-sortal identities of these new objects (Julius Caesar problem).
Issue of *inconsistent* objects: courses of values (*Wertverläufe*).
Possible solution: relax requirement that truth-values of *all* identities must be settled in order to introduce new terms/objects.

3) Frege's 3-world ontology:

This is a division of *Bedeutungen*, since what there is can be referred to (including senses, by indirect discourse and semantic discourse, and subjective episodes and states). Issues unresolved by Frege's "third-world realism" about senses, objective but not actual (=wirklich, cf. *werken*, to work), which includes *facts* (for Frege: "thoughts that are true") include status of

- i. *concepts*, which are *Bedeutungen* of functions whose values are truth-values, and
- ii. *Truth values*,
- iii. *Functions*, apart from *sense* functions, which are presumably in the world of senses.
- iv. *abstracta* that are not *linguistic*, such as the direction of the Earth's axis, or its center of mass.

These are certainly invoked in *causal explanations*. They are "idealizations" in the sense Nancy Cartwright diagnoses in *How the Laws of Physics Lie*.

These all raise the issue of whether the world of *actual* objective things, contains more than *objects*. Remember, the *facts* are in the world of *senses*—for instance, the *fact* that an object has a certain property or stands in a certain relation, that is, that a *concept* applies to it.

This is one transition to consideration of Sellars's *nominalism*—in particular, his *metalinguistic nominalism*.

4) Sellars's two-world ontology:

Discourse-dependence rather than subject (mind)-dependence.

Nominalistic, in the sense of restricted to *objects*: no functions or (therefore) concepts, properties, relations, propositions or (therefore) facts.

Nominalism is the result of Sellars's (following Carnap's) *metalinguistic* analysis of Fregean *concepts*: properties and relations, and the *propositions* and so (non-Fregean) *facts*.

5) Sense-dependence vs. reference-dependence:

The sense in which the species of *abstracta* that includes properties (=Fregean *concepts*), relations, and propositions (including Fregean *facts*) are *metalinguistic* need involve only *sense-dependence*, not *reference-dependence*. It is *not*, as Sellars takes it to be, an argument for *nominalism*.